COMPUTER ENGINEERING

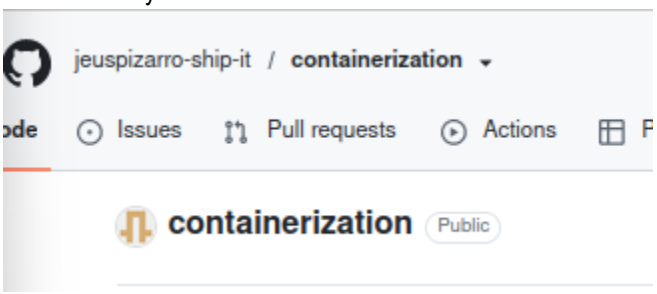
QUEZON CITY

COURSE CODE/TITLE:	CpE212	**SECTION:**	CpE31s2
DATE PERFORMED:	10/15/26	**DATE SUBMITTED:**	10/15/26
NAME:	Jeus Guille V. Pizarro	**INSTRUCTOR:**	Engr Jimlord Quejado

ACTIVITY NO. 11 **Containerization**

1. PROCEDURE OUTPUT

1. Create a new repository for this activity.



GitHub repository created for Activity No. 11 named 'containerization'.

2. Install Docker and enable the docker socket.

```
pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/containerization$ sudo systemct
 status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset
     Active: active (running) since Thu 2026-10-15 08:47:44 PST; 1min 8s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 5221 (dockerd)
      Tasks: 11
     Memory: 26.1M
        CPU: 3.615s
     CGroup: /system.slice/docker.service
             └─5221 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/cont
```

Docker service status output showing docker.service is active (running) and docker.socket is enabled.

3. Add to Docker group to your current user.

```
pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/containerization$ sudo usermod -
aG docker $USER
pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/containerization$ newgrp docker
pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/containerization$
```

Command output showing the current user added to the 'docker' group using usermod and refreshed using newgrp.

```
pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/containerization$ docker run hel
lo-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/
```

Successful execution of 'docker run hello-world' verifying that the Docker daemon and client installation are operational.

Verified if docker is running

4. Create a Dockerfile to install web and DB server.

```
FROM ubuntu:latest


ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y nginx mariadb-server maria
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD service mariadb start && nginx -g 'daemon off
```

Dockerfile configuration installing Nginx web server and MariaDB database server on an Ubuntu base image with exposed ports 80 and 3306.

5. Install and build the Dockerfile using Ansible.

```
TASK [Install Docker and Pip (CentOS/RHEL)] ************************************
skipping: [pizza3]
changed: [pizza_centos]

TASK [Install Docker Python SDK via pip (CentOS/RHEL)] *************************
skipping: [pizza3]
changed: [pizza_centos]

TASK [Ensure Docker service is started] ****************************************
ok: [pizza3]
fatal: [pizza_centos]: FAILED! => {"changed": false, "msg": "Could not find the
requested service docker: host"}

TASK [Create project directory] ************************************************
ok: [pizza3]

TASK [Copy Dockerfile to remote host] ******************************************
ok: [pizza3]

TASK [Build Docker Image] ******************************************************
changed: [pizza3]
```

```
TASK [Build Docker Image] ******************************************************
changed: [pizza3]

TASK [Ensure container is running] *********************************************
changed: [pizza3]

PLAY RECAP *********************************************************************
pizza2                     : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
pizza3                     : ok=7    changed=2    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
pizza_centos               : ok=3    changed=2    unreachable=0    failed=1    s
kipped=1    rescued=0    ignored=0
workstation1               : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0

pizza1@pizza1-local:~/CPE212PIZARRO/ansible-lab/containerization$ ss
```

Ansible playbook execution recap showing successful tasks run against target hosts.

```yaml
---
- name: Install, Build, and Deploy Docker Contain
  hosts: localhost
  connection: local
  become: yes

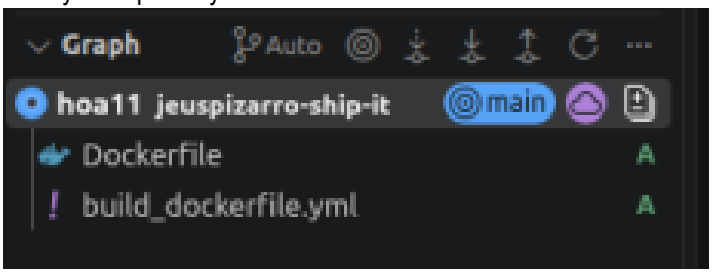
  tasks:
    - name: Ensure Docker Python SDK is installed
      apt:
        name: python3-docker
        state: present

    - name: Build Docker Image from Dockerfile
      community.docker.docker_image:
        build:
          path: .
        name: web-db-app
        tag: latest
        source: build

    - name: Create and start container from built image
      community.docker.docker_container:
        name: web_db_container
        image: web-db-app:latest
        state: started
        ports:
          - "8080:80"
          - "3306:3306"
```

Ansible YAML playbook 'build_dockerfile.yml' automating Docker Python SDK installation, Docker image building, and container deployment.

6. Add, commit and push it to your repository.



Git repository commit graph displaying 'Dockerfile' and 'build_dockerfile.yml' added and committed under branch main.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:
Through this activity, containerization was successfully implemented and automated using Docker and Ansible. By building lightweight images from Dockerfiles and deploying web/database infrastructure, container technology simplifies deployment, isolates runtime environments, and provides scalable orchestration for modern software development.

1. What are the benefits of implementing containerizations?
Containerization ensures reliable, consistent deployments and resource efficiency by sharing the host OS kernel instead of using traditional virtualization overhead. Additionally, it enables rapid scalability and secure isolation through lightweight environments that integrate seamlessly into continuous delivery pipelines

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of code**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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